

CarbonBridge

An Intelligent B2B CO₂ Marketplace & Transaction Platform

TEAM NAME
AESTRO

PROBLEM STATEMENT
PS8: Carbon Capture-to-Product Matchmaking

SUBMISSION ROUND
Ideation Round

One-Line Solution

CarbonBridge connects point-source industrial emitters capturing CO₂ with commercial utilization off-takers—matching multi-seller lots, streamlining dynamic pricing and quality attestation, and tracking physical custody from allocation to final delivery.

Why It Matters

- **Transforms Fragmented Supply:** Converts isolated, on-site captured CO₂ volumes into a transparent, liquid digital marketplace.
- **Solves Quantity Mismatches:** Pools smaller lots from multiple regional capture plants to satisfy large single-buyer off-take demands.
- **Flexible Commercial Mechanisms:** Provides emitters direct Fixed-Price listings or dynamic Seller-Side Auctions to maximize yield.
- **Guaranteed Transaction Integrity:** Unique Batch IDs and rigorous balance validation prevent accidental over-allocation or phantom stock.

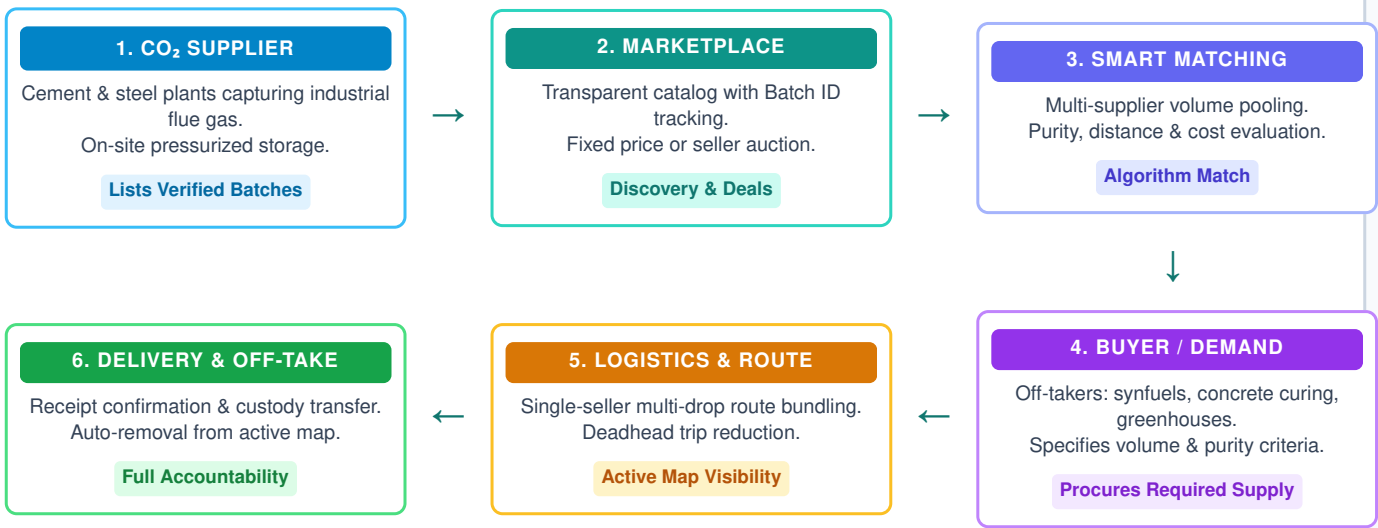


DIAGRAM 1 — HIGH-LEVEL SYSTEM OVERVIEW (SPACIOUS 2-ROW LAYOUT)

PROBLEM UNDERSTANDING

Critical friction points and structural failures in the Carbon Capture-to-Product ecosystem

CONTEXT & CHALLENGE

PS8 addresses a vital decarbonization bottleneck: while point-source emitters increasingly capture carbon, they lack a dedicated commercial distribution infrastructure. Captured gas remains stranded on-site due to uncoordinated bilateral negotiations, manual discovery, and logistics opacity.

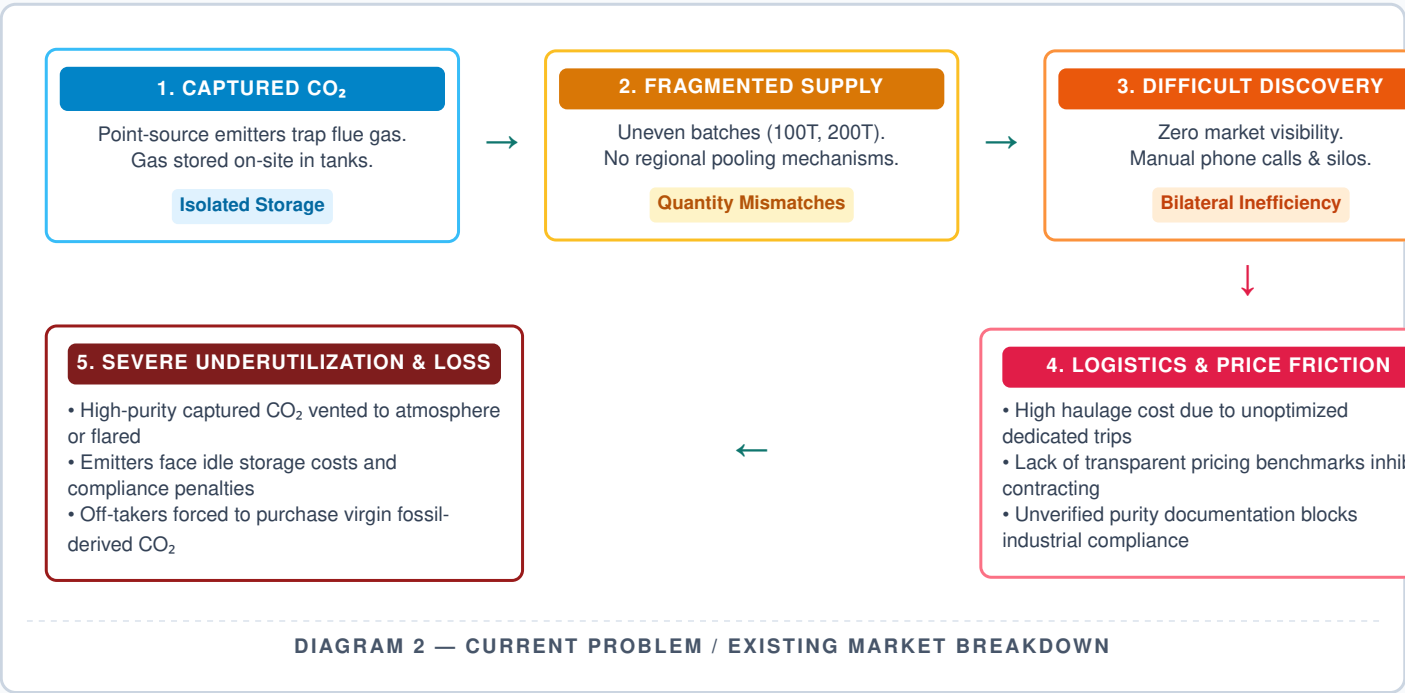
Supply-Side Reality (Emitters)

Cement, steel, chemicals, and energy plants capture carbon to comply with mandates or voluntary goals. However, lacking marketing channels, CO₂ is either vented back or accumulated in costly pressurized holding tanks without reliable monetisation.

Demand-Side Reality (Commercial Off-Takers)

Manufacturers of synthetic aviation fuels (e-fuels), carbonated precast concrete, and high-yield commercial greenhouses require reliable bulk CO₂ but struggle to source consistent volume and purity specifications from local industrial clusters.

MARKETPLACE GAP	MANIFESTATION & SYSTEMIC IMPACT
Visibility Gap	No centralized registry exists; suppliers and off-takers operate in information silos, relying on sporadic broker phone calls.
Quantity Mismatch	Individual capture batches (e.g., 80–150T) rarely match off-taker lot requirements (500T+), stalling transaction execution.
Purity & Quality Gap	Industrial buyers cannot audit gas purity certifications upfront, delaying contracting due to contamination liability fears.
Geographic Friction	CO ₂ transport requires pressurized cryogenic tankers; without route aggregation, transport freight exceeds raw gas value.
Price Opacity	Absence of transparent spot or benchmark pricing leads to arbitrary pricing and protracted bilateral bargaining.
Transaction Integrity	Informal allocations create risk of "phantom supply" where one pressurized storage lot is committed to multiple buyers.



The CarbonBridge Platform

CarbonBridge transforms uncoordinated, ad-hoc CO₂ sales into an automated digital trading exchange. By marrying algorithmic multi-supplier matching with real-time inventory integrity and transaction-based logistics tracking, the platform enables industrial off-takers to procure CO₂ as easily as any standardized bulk commodity.

Algorithmic Demand Fulfillment

Rather than leaving buyers to manually piece together fragmented supply lots, CarbonBridge's multi-supplier engine evaluates spatial proximity, purity thresholds, and landed logistics expenses to generate optimized composite fulfillment proposals.

Guaranteed Physical Custody

Every listed metric ton is anchored to an immutable Batch ID. Mathematical allocation gating enforces that commitments never exceed verified residual balance, eliminating double-selling risks and securing audit readiness.

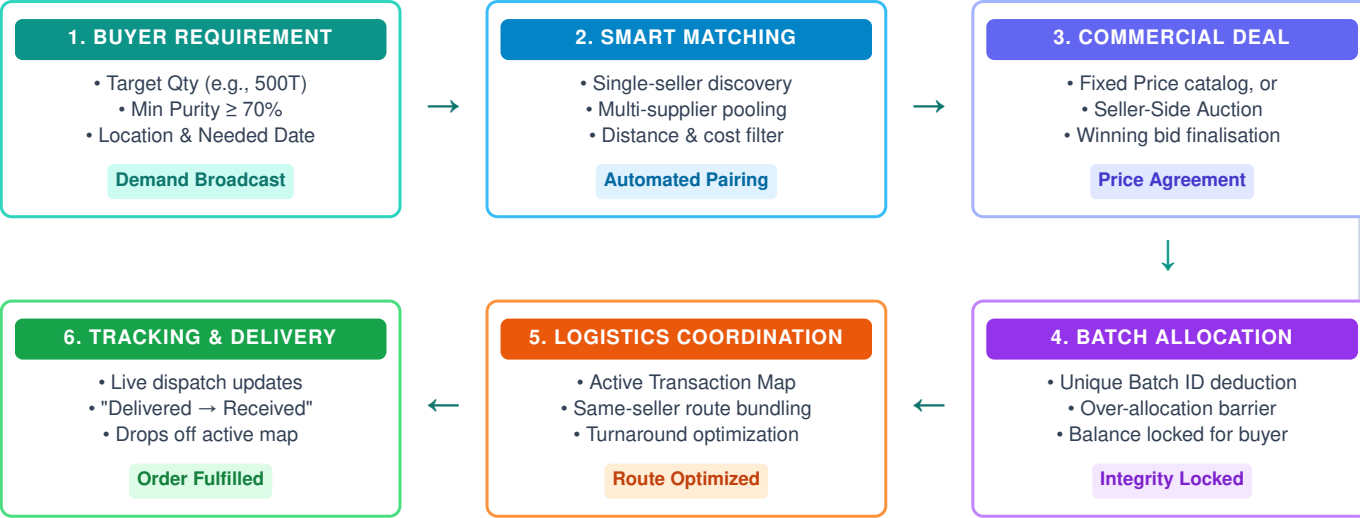


DIAGRAM 3 — END-TO-END SOLUTION WORKFLOW (2-ROW CIRCULAR FLOW)

KEY FEATURES: MARKETPLACE & MATCHING

Granular requirement ingestion, dual selling modalities, and multi-seller lot pooling

CORE ENGINE

1. Buyer Requirement Specification

Buyers configure precise industrial demand parameters:

- **Volume & Quality:** Required tonnage (T) and min purity threshold (e.g., ≥70%).
- **Temporal & Location:** Plant delivery coordinates and strict target arrival dates.
- **Commercial Bounds:** Upper budget ceiling and intended application profile.

2. Seller Listing & Trading Modalities

Sellers list captured gas with Batch ID, volume, purity, and location:

- **Fixed Price:** Direct listing at set rate (e.g., ₹2,500/T) for immediate procurement.
- **Seller-Side Auction:** Seller sets base reserve (e.g., ₹2,500/T) and window (e.g., 7 days). Multiple buyers compete via upward bidding; seller finalises winner.

Smart Multi-Supplier Matching Mechanics

Industrial buyers frequently require large bulk quantities (e.g., 500T) exceeding any single emitter's storage. Rather than rejecting orders, the matching engine aggregates complementary seller lots into a unified package based on: **Purity Compatibility** (all ≥ buyer min), **Geographic Proximity**, **Delivery Timeline Feasibility**, and **Landed Unit Cost**.



DIAGRAM 4 — MULTI-SUPPLIER MATCHING ENGINE (VOLUME POOLING & CO-OPTIMIZATION)

KEY FEATURES: PRICE GUIDANCE & BATCH INTEGRITY

Transparent price benchmarks, quality attestation, and strict allocation protection

TRUST & GOVERNANCE

Transparent Recommended Pricing

Separate from dynamic bidding, the platform computes a deterministic benchmark range (e.g., ₹2,400–₹2,700/T). Grounded in verified purity levels, batch size, cluster logistics benchmarks, and regional supply-demand ratios, sellers can freely adopt, adjust, or bypass this recommendation.

Certificate of Analysis (CoA) Visibility

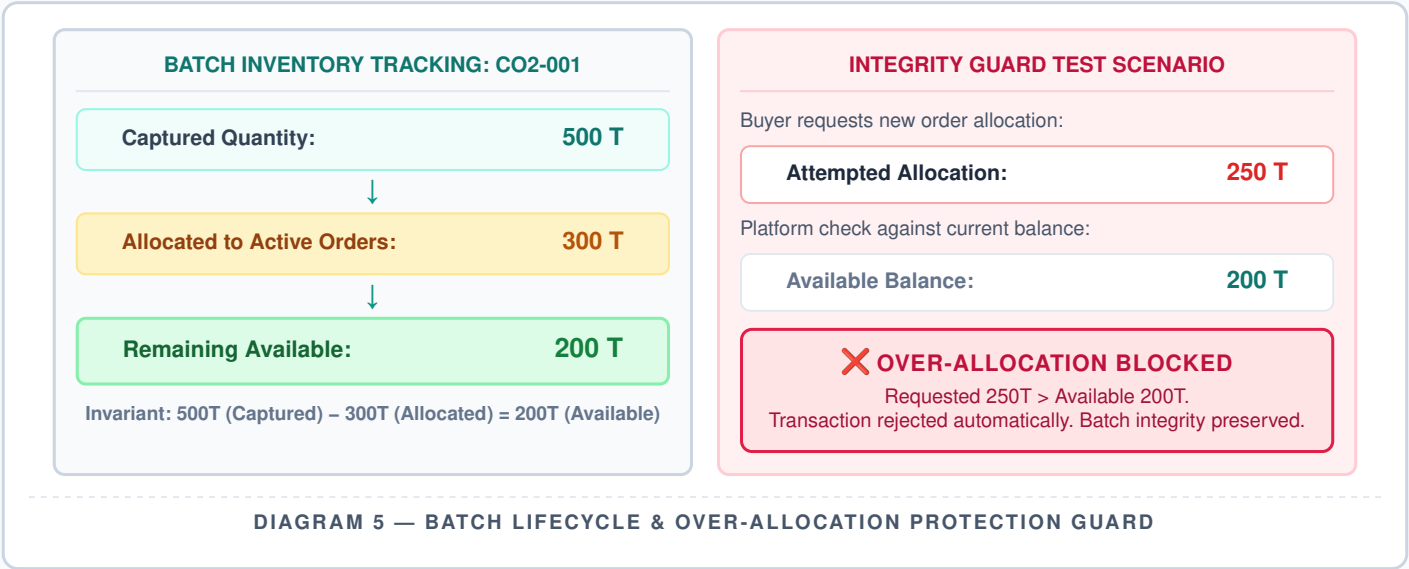
Suppliers upload third-party lab assay reports detailing purity, moisture, and trace hydrocarbons. To maintain absolute operational honesty, listings display "Certificate Uploaded" or "Certificate Available" for off-taker inspection, avoiding unverified claims of platform certification.

Batch ID & Over-Allocation Protection Mechanism

Every physical batch is assigned an immutable Batch ID. The core engine continuously enforces the strict invariant formula:

Available Quantity = Captured Quantity – Allocated Quantity

If an attempted allocation exceeds the current available balance, the platform hard-blocks the transaction, totally eliminating double-allocation and phantom inventory.



KEY FEATURES: LOGISTICS & TRACKING

Single uncluttered transaction map, same-seller route consolidation, and end-to-end custody tracking

FULFILLMENT OPS

Transaction-Based Map Architecture

CarbonBridge provides **one single, focused map** strictly scoped to the active transactions of the authenticated enterprise.

- **Seller View:** Pinpoints buyers with active, allocated, undelivered lots.
- **Buyer View:** Pinpoints source capture plants dispatching allocated supply.
- **Zero Clutter:** Upon final "Received" state, records cleanly drop off the active map.

Same-Seller Multi-Buyer Route Bundling

When an emitter fulfills allocations for multiple buyers along contiguous transport corridors, the platform flags consolidation opportunities:

- **Strict Boundary:** Evaluates orders from ONE seller to multiple regional buyers (never combining distinct sellers).
- **Deadhead Elimination:** Multi-drop dispatch runs significantly decrease empty-run mileage and cryogenic freight overhead.

SAME-SELLER MULTI-DROP ROUTE OPTIMIZATION VISUAL



TRANSACTION MAP VISIBILITY RULE

- Shows **ONLY** active, allocated, undelivered runs.
- Drops completely once order reaches "Received".
- Maintains low latency and high regional usability.

STANDARDIZED ORDER & CUSTODY LIFECYCLE

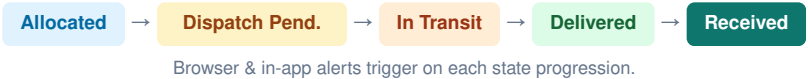


DIAGRAM 6 — TRANSACTION MAP, SAME-SELLER ROUTE CONSOLIDATION & ORDER LIFECYCLE

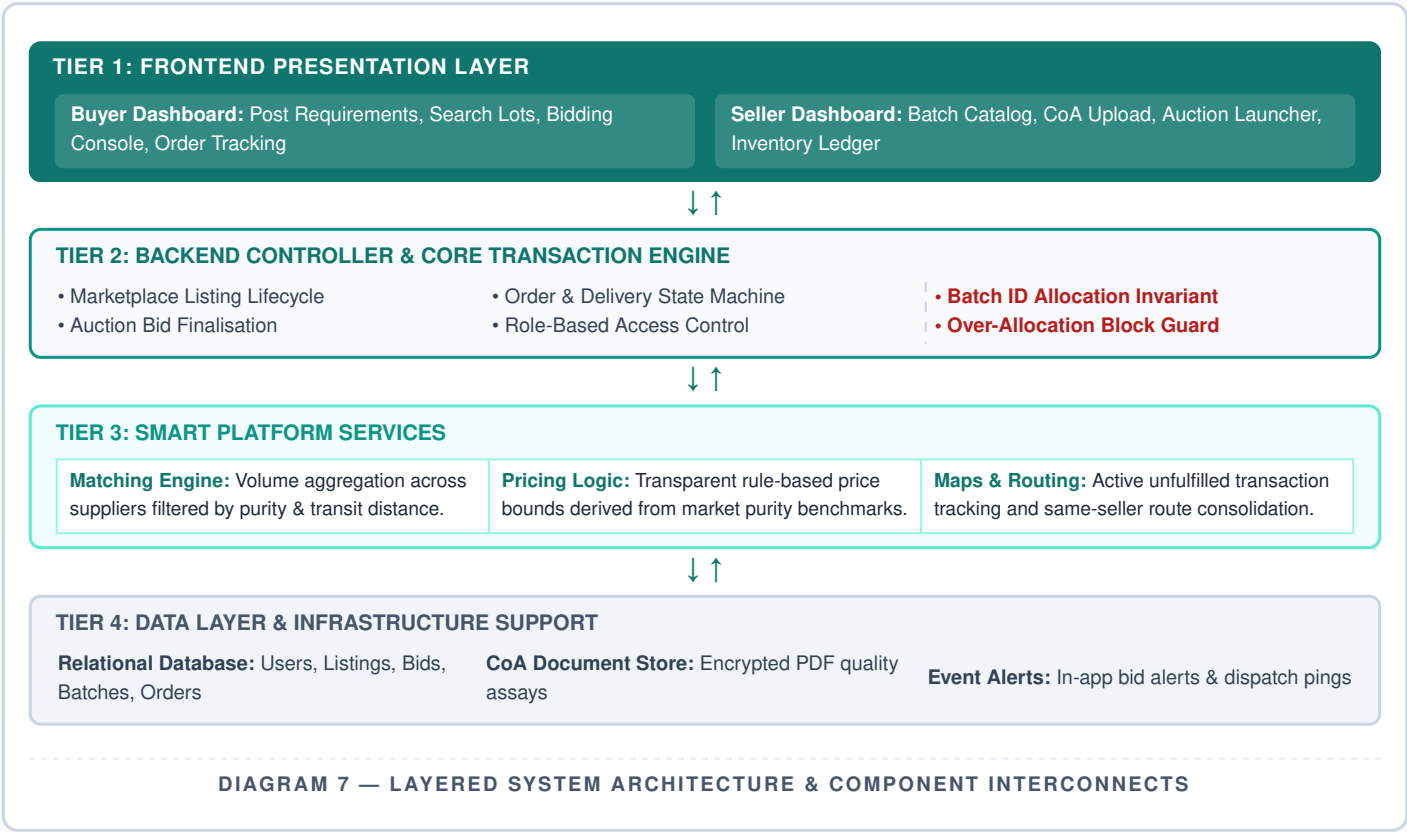
SYSTEM APPROACH: HOW IT WORKS

Layered, modular system architecture engineered for rapid PoC deployment and enterprise scalability

TECHNICAL ARCHITECTURE

The platform architecture is designed to be cleanly buildable by a focused engineering team within hackathon constraints. Where real-world physical telemetry is unavailable, CarbonBridge uses deterministic rule-based algorithms and structured relational schemas to guarantee operational robustness.

SYSTEM LAYER	ARCHITECTURAL ROLE & CORE RESPONSIBILITIES
Frontend Dashboards	Role-based UI portals for Buyers (requirements, bids) and Sellers (batch listings, inventory management, auction setup).
Backend Controller	REST API handling transactions, listing state machines, bid management, and the Over-Allocation Integrity Guard.
Smart Services	Multi-supplier combinatorial matching engine, rule-based price indexer, and spatial route consolidation detector.
Data & Support	PostgreSQL relational datastore for transactions/lots, document store for CoA PDFs, and WebSocket/in-app notification bus.



BUSINESS VALUE & REVENUE MODEL

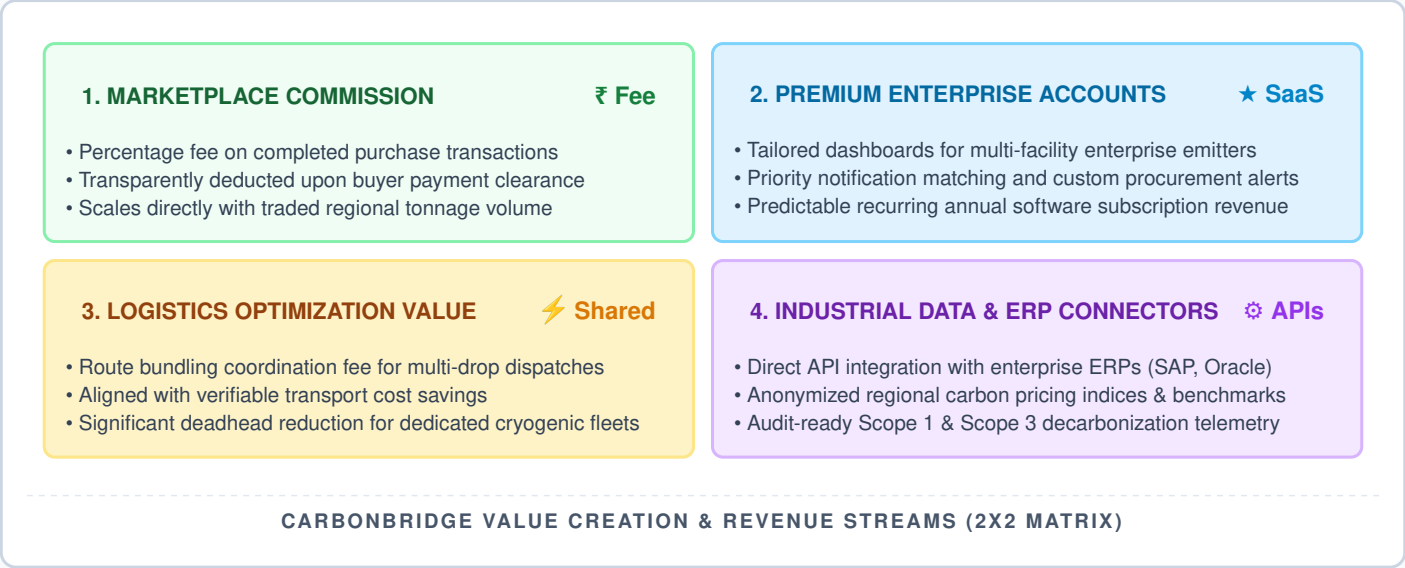
Multi-stakeholder value creation, commercial sustainability, and revenue diversification

COMMERCIAL VIABILITY

STAKEHOLDER	TANGIBLE PLATFORM VALUE DELIVERED
CO ₂ Suppliers (Emitters)	Monetises trapped flue gas; provides transparent price discovery, auction premiums, and rigorous batch inventory control.
Industrial Buyers (Off-Takers)	Discovers multi-supplier fulfillment for large volumes; provides CoA visibility and reliable delivery timelines.
Logistics Operators	Achieves higher tanker utilization via same-seller multi-buyer route consolidation, decreasing empty-mile deadheading.
Environmental Impact	Diverts captured carbon into durable products (synfuels, precast concrete), fostering a practical circular carbon economy.

Business & Monetisation Model

Core Revenue Engine: CarbonBridge operates on a predictable, volume-scaled marketplace transaction commission fee applied to successfully executed and settled CO₂ sales. Long-term commercialization introduces tiered enterprise SaaS models and API integrations.



MVP SCOPE & STRATEGIC ROADMAP

Practical demonstration boundaries, hackathon deliverables, and future expansion vision

EXECUTION PLAN

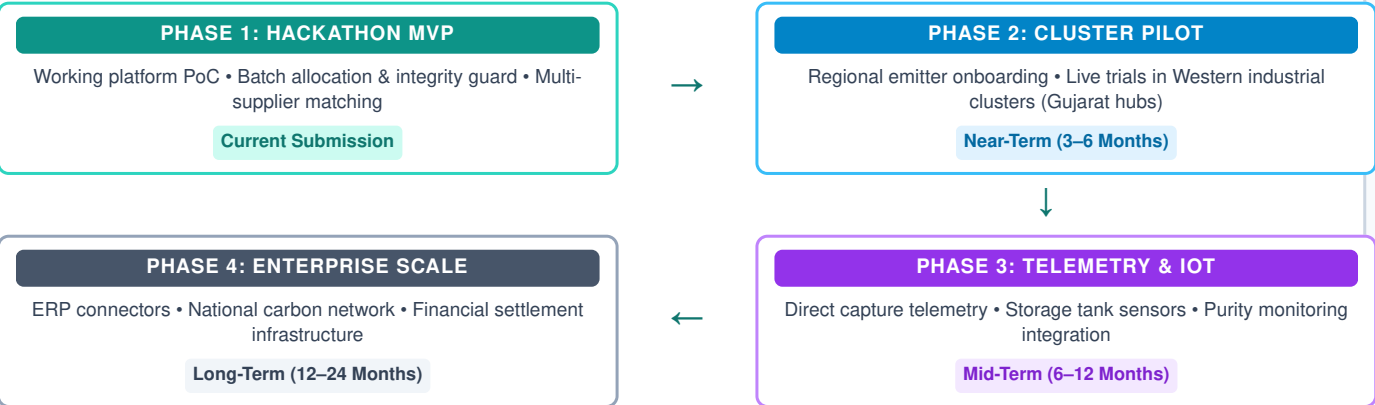
Hackathon MVP — Core Deliverables

- **Buyer Ingestion:** Post volume, min purity, location, date needed.
- **Seller Catalog:** Batch listing with unique Batch ID & fixed pricing.
- **Multi-Supplier Matching:** Algorithmic volume aggregation across lots.
- **Trading Engine:** Fixed-price procurement & seller-side auction.
- **Allocation Guard:** Balance deduction & hard over-allocation block.
- **Active Map & Tracking:** Live lifecycle progression visualization.
- **Demo Extras:** CoA viewing, recommended pricing bounds, alerts.

Future Roadmap — Commercial Scale

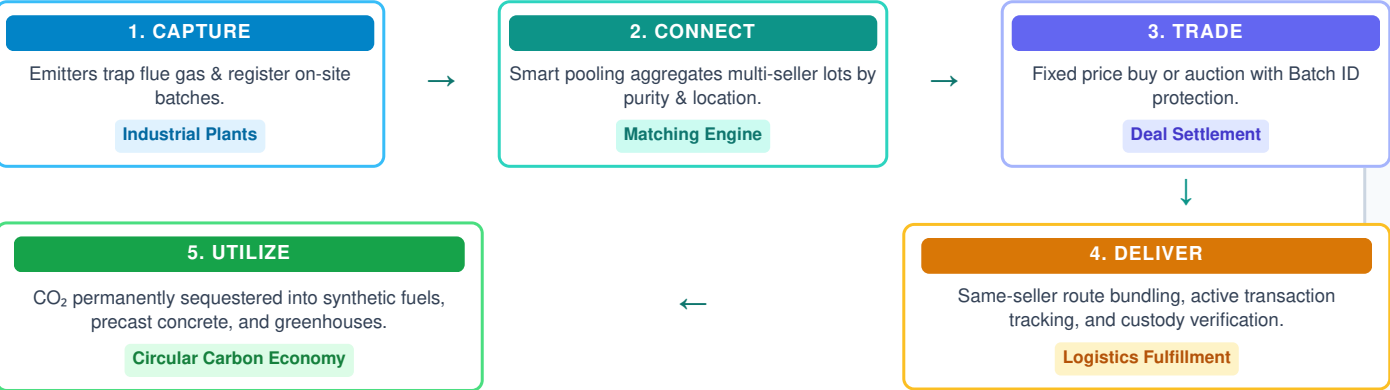
- **Automated IoT Telemetry:** Direct capture tank sensor feeds.
- **AI Predictive Pricing:** Dynamic spot index & demand forecasting.
- **Assay Verification:** Third-party accredited digital CoA certification.
- **Carrier Integration:** API dispatch feeds with dedicated cryo fleets.
- **Enterprise ERP Connectors:** Automated SAP/Oracle supply ingestion.
- **Multi-Cluster Expansion:** Scaling across national industrial corridors.

DELIVERY PHASING & EXPANSION TRAJECTORY



STRATEGIC HORIZON & MATURITY PATHWAY

THE MISSING MARKETPLACE LAYER FOR CCUS: FULL LIFECYCLE INTEGRATION



FINAL VISUAL — CAPTURE • CONNECT • TRADE • DELIVER • UTILIZE